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Final Project

Through the course of this project, it became clear that GenAI is best suited for evaluating reasoning ability rather than solving problems directly. My initial reflections on issues faced in the process of developing an AI system were superficial, often limiting themselves to symptomatic observations like "the model's accuracy decreased" or "the model crashed." When I fed these reflections into GenAI, I found that it usually classified them as "Bad" and replied using Socratic questioning techniques instead of offering solutions.

As a result, I was able to think about the problem in greater depth. For instance, in one situation where there was an unexpected decrease in the model's accuracy, I had initially assumed that the model architecture was insufficient. But after reflecting the matter again, I discovered that the issue stemmed from my having inadvertently shuffled the labels of the training dataset separately from the features, disrupting their correspondence.

The final lesson to take away from the course is that the answers produced by GenAI are not what makes it valuable as a tool. The ability of the AI to evaluate the reasoning quality and ask the right questions enabled me to use it as a cognitive aid that helped improve my problem-solving skills better than its capability to serve as a debugger did.

One of the crucial lessons that this project taught me is that AI models are very sensitive and prone to making mistakes that do not lead to error messages and may go unnoticed. As compared to ordinary software, an AI system may run perfectly while delivering the wrong answer without notifying the user about any problems.

For instance, while working on one of my projects, I committed a serious mistake by using the feature scaler on the entire database rather than the training data only. As a result, I achieved higher accuracy scores but, in fact, got misleading results since my test database was somehow incorporated into the training process.

I also realized that maintaining consistent behavior between different components of the system was very important. Problems such as inconsistent behavior between label encoding and loss function, inappropriate output layers' dimension, and using the wrong activation function could lead to system failure or significant reduction in its performance. This demonstrated to me that developing AI solutions is not only an issue of developing models but also making sure that all of their components are aligned with each other.

In summary, I have learned that debugging AI systems involves systematic identification of problems, which should include knowledge of why they occurred. The key skill in debugging any kind of system lies in figuring out why the problem occurred in the first place.

Working alongside GenAI as an intelligence ally enhanced my problem-solving capabilities. I found out that good development of AI technology requires proper data management, a good system design, and an understanding of cause and effect in the AI process.